

REPORT PROJECT

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1-Project Objective

The primary objective of this project is to computationally model and validate the turbulent deposition of aerosol particles in a 90° pipe bend using ANSYS Fluent. Standard stochastic tracking methods within the standard Discrete Phase Model (DPM) often fail to accurately predict deposition rates for low-inertia particles due to the artificial accumulation of particles in near-wall boundary layers. To resolve this non-physical behavior, this project implements the drift-corrected Continuous Random Walk (CRW) model proposed by Dehbi (2008) through the integration of a custom User-Defined Function (UDF).

The numerical investigation is designed to capture both deposition mechanisms present in curved geometries: turbulent dispersion, which governs the behavior of small particles, and inertial impaction, which dictates the trajectory of larger particles.

The specific scope of the computational work encompasses:

- **Flow Field Resolution:** Simulating the continuous fluid phase using the Reynolds Stress Model (RSM) to accurately capture the highly anisotropic turbulence and secondary flow structures inherently generated within the 90° bend.
- **Targeted Mesh Optimization:** Conducting a rigorous mesh independence study focused explicitly on the viscous sublayer. Maintaining a wall $y^+ \approx 1$ and utilizing a high-density inflation layer configuration to ensure the precise spatial calculation of Reynolds stress gradients. This step is critical, as these gradients mathematically drive the UDF's drift-correction term.
- **Custom UDF Integration:** Replacing standard turbulent dispersion tracking with a customized Langevin equation (DEFINE_DPM_BODY_FORCE) that actively corrects the fluctuating velocity components of the dispersed phase.
- **Benchmark Validation:** Comparing the final simulated deposition fractions against the empirical correlation established by Pui et al. (1987) and computationally reproducing the validated deposition curve (Figure 8) presented in Dehbi (2008).

Ultimately, this project aims to demonstrate a robust, grid-independent methodology within ANSYS Fluent 2024 R2 capable of accurately predicting complex multiphase aerosol deposition across a wide spectrum of particle sizes.

2. Geometry

The computational domain represents a 90° pipe bend, constructed to strictly replicate the experimental conditions of Pui et al. (1987), which serves as the benchmark for validation in this study.

The primary geometric parameters of the bend are defined by the pipe diameter D , and the radius of curvature R . To match the experimental curvature ratio defined as $R/r = 4$ (where r is the pipe radius), the computational model utilizes a pipe diameter of $D = 2 \text{ cm}$ ($r = 1 \text{ cm}$) and a corresponding radius of curvature of $R = 4 \text{ cm}$.

To ensure the numerical accuracy of the flow physics entering and exiting the bend, straight pipe extensions were incorporated into the domain:

- **Upstream (Inlet) Section:** A straight pipe segment of length L ($5D$ or 10 cm) is added before the bend. This extension is physically necessary to allow the turbulent velocity profile to become fully developed before interacting with the curvature, ensuring the bend entrance is free from non-physical inlet boundary disturbances.
- **Downstream (Outlet) Section:** A straight pipe segment of length L ($5D$ or 10 cm) is added after the bend. This region prevents reverse flow at the pressure outlet boundary from artificially influencing the secondary flow structures developing within and immediately after the bend.



Figure 1 : A 3D CAD schematic

3. Mesh Generation & Independence Study

Generating a high-quality computational grid was a critical step in ensuring the numerical stability and physical accuracy of the drift-corrected Continuous Random Walk (CRW) model. The geometry was discretized using a structured hexahedral mesh, with a final cell count of approximately 5.3 .

3.1 Near-Wall Treatment and Spatial Resolution

Accurate prediction of turbulent aerosol deposition necessitates stringent control over the near-wall mesh architecture. Because the Reynolds Stress Model (RSM) with Enhanced Wall Treatment was employed to resolve the anisotropic turbulence within the 90° bend, the boundary layer mesh was explicitly designed to fully resolve the viscous sublayer.

To achieve this, the first cell height (70 micron) adjacent to the wall was precisely calculated to yield a non-dimensional wall distance of $y^+ \approx 0.95$. Furthermore, a high-density inflation layer strategy was implemented, utilizing 30 prismatic layers with a strict growth rate of 1.1. This ensured smooth volume transitions and captured the steep velocity gradients present in the near-wall region.

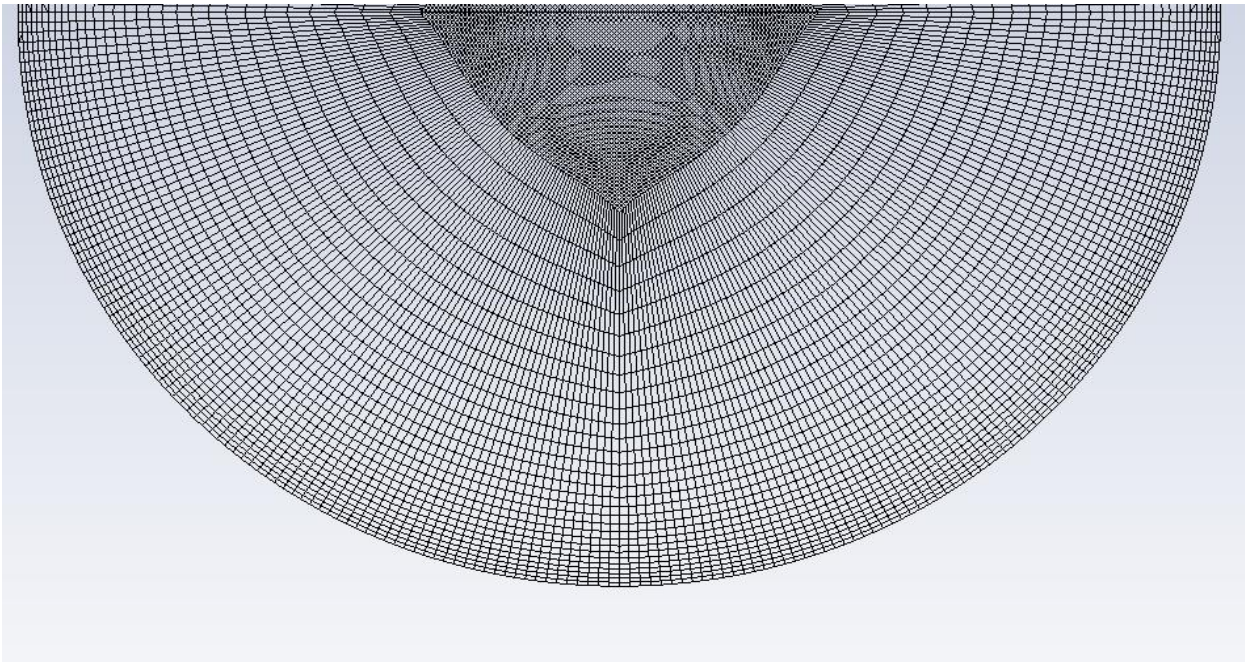


Figure 2 : A zoomed-in cross-sectional (from front) view of the mesh at the bending section, displaying the dense inflation layers and smooth transition to the bulk flow mesh.(5.3 million)

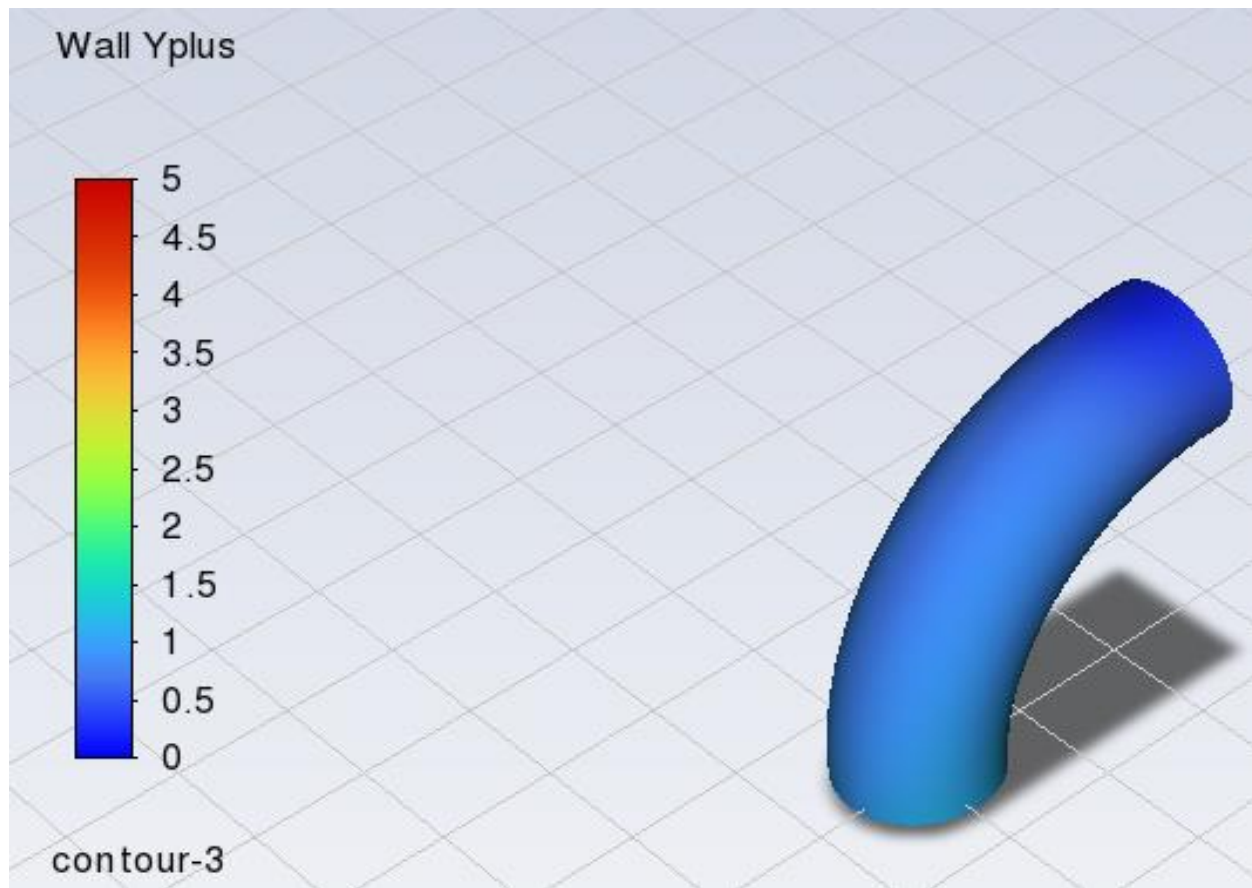


Figure 3 : A contour plot of the wall y^+ values across the 90° bend, demonstrating that $y^+ < 1$ is maintained throughout the critical deposition zones.

3.2 Mesh Independence and Low-Inertia Particle Sensitivity

During the verification phase, a rigorous mesh independence study revealed a critical grid sensitivity specific to low-inertia particles ($d_p = 5$ and $10 \mu\text{m}$). While the baseline mesh (about 2.5 million cells) adequately predicted the deposition fractions for larger particles, whose trajectories are primarily governed by macroscopic inertial impaction and mean flow structures, it produced significant deviations from benchmark data for the smaller particles.

The root cause of this discrepancy lies in the physics of turbulent dispersion. The deposition of $5\text{-}10 \mu\text{m}$ particles is dominated by their interaction with turbulent eddies in the immediate vicinity of the wall. To prevent non-physical accumulation, the custom UDF applies a drift correction term driven by the spatial gradients of the Reynolds stresses.

By refining the mesh locally within the bend and severely increasing the node density within the $y^+ < 100$ region, the computation of the Reynolds stress gradients was stabilized.

This targeted refinement successfully resolved the discrepancy for the 5 and 10 μm particles, achieving true mesh independence across the entire spectrum of simulated particle sizes.

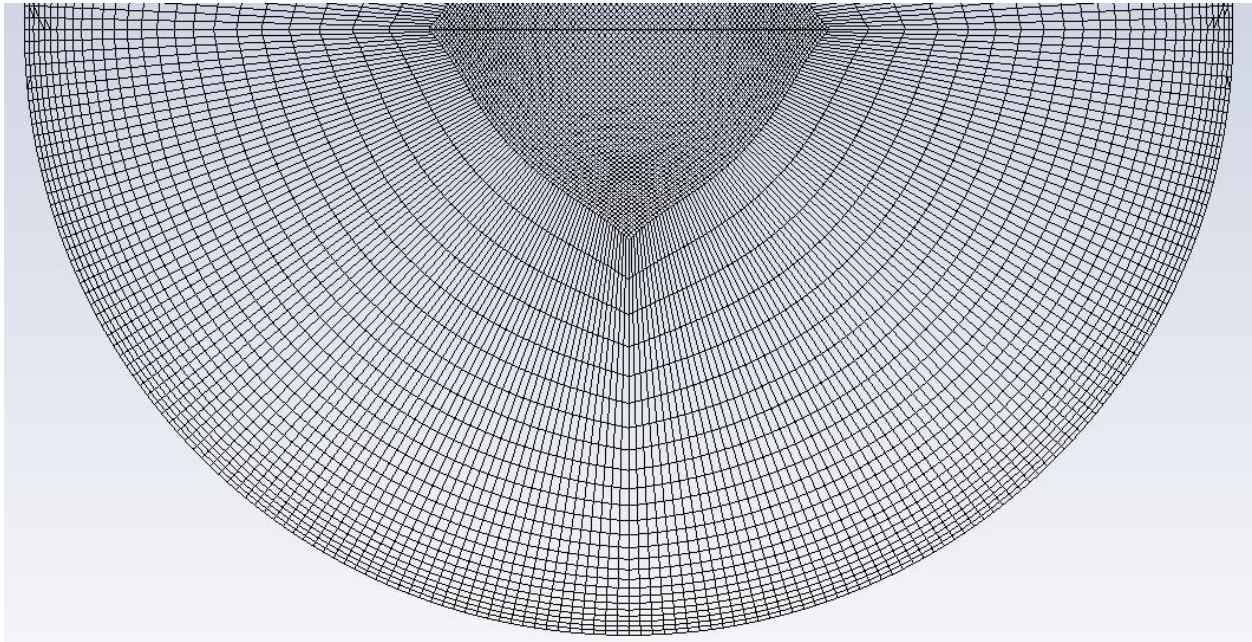


Figure 4 (Mesh independent) : A zoomed-in cross-sectional (from front) view of the baseline mesh at the bending section, it produced significant deviations from benchmark data for the small particles. (about 2.5 million cells)

4. ANSYS Fluent Setup & Physics Modeling

The numerical simulation was conducted using the commercial CFD solver ANSYS Fluent 2024 R2, utilizing a pressure-based, steady-state solver.

The simulation was conceptually divided into two phases: resolving the continuous fluid phase (Eulerian framework) and tracking the dispersed aerosol particles (Lagrangian framework).

Given the low volume fraction of the aerosol particles, a one-way coupling approach was assumed, meaning the fluid flow dictates particle trajectories, but the particles do not affect the fluid flow.

4.1 Continuous Phase and Boundary Conditions

The working fluid was defined as air at standard conditions (density = 1.225 kg/m^3 , viscosity = 1.81×10^{-5}).

The boundary conditions were established to replicate the Pui et al. (1987) benchmark:

- **Inlet:** A Velocity Inlet boundary condition was applied with a uniform bulk velocity of 3.0 m/s . This corresponds to a turbulent Reynolds number of $Re = 4080$ based on the pipe diameter 2 cm .
- **Outlet:** A Pressure Outlet boundary condition was specified with a gauge pressure of 0 Pa .
- **Walls:** The pipe walls were modeled as stationary with a standard no-slip condition.

For spatial discretization, the SIMPLE algorithm was utilized for pressure-velocity coupling. To minimize numerical diffusion, Second-Order Upwind schemes were enforced for the spatial discretization of momentum and all turbulence quantities.

4.2 Turbulence Modeling

To accurately capture these complex flow physics, the **Reynolds Stress Model (RSM)** was employed.

By solving transport equations for each individual Reynolds stress component, the RSM directly accounts for the effects of streamline curvature and swirl.

Furthermore, **Enhanced Wall Treatment** was enabled to resolve the flow perfectly down to the viscous sublayer ($y^+ \approx 1$), which is a strict mathematical prerequisite for the drift-correction UDF.

Viscous Model

Model

- ☐ Inviscid
- ☐ Laminar
- ☐ Spalart-Allmaras (1 eqn)
- ☐ k-epsilon (2 eqn)
- ☐ k-omega (2 eqn)
- ☐ Transition k-kl-omega (3 eqn)
- ☐ Transition SST (4 eqn)
- ☒ Reynolds Stress (7 eqn)
- ☐ Scale-Adaptive Simulation (SAS)
- ☐ Detached Eddy Simulation (DES)
- ☐ Large Eddy Simulation (LES)

Reynolds-Stress Model

- ☒ Linear Pressure-Strain
- ☐ Quadratic Pressure-Strain
- ☐ Stress-Omega
- ☐ Stress-BSL

Reynolds-Stress Options

- ☒ Wall BC from k Equation
- ☒ Wall Reflection Effects

Near-Wall Treatment

- ☐ Standard Wall Functions
- ☐ Scalable Wall Functions
- ☐ Non-Equilibrium Wall Functions
- ☒ Enhanced Wall Treatment
- ☐ User-Defined Wall Functions

Enhanced Wall Treatment Options

- ☒ Pressure Gradient Effects

Model Constants

Cmu: 0.09

C1-Epsilon: 1.44

C2-Epsilon: 1.92

C1-PS: 1.8

C2-PS: 0.6

C1'-PS: 0.5

C2'-PS: 0.3

TKE Prandtl Number: 1

TDR Prandtl Number: 1.3

OK Cancel Help

Solution Methods

Pressure-Velocity Coupling

Scheme: SIMPLE

Flux Type: Rhie-Chow: momentum based ☒ Auto Select

Spatial Discretization

Gradient: Least Squares Cell Based

Pressure: Second Order

Momentum: Second Order Upwind

Turbulent Kinetic Energy: Second Order Upwind

Turbulent Dissipation Rate: Second Order Upwind

Reynolds Stresses: Second Order Upwind

Pseudo Time Method

Off

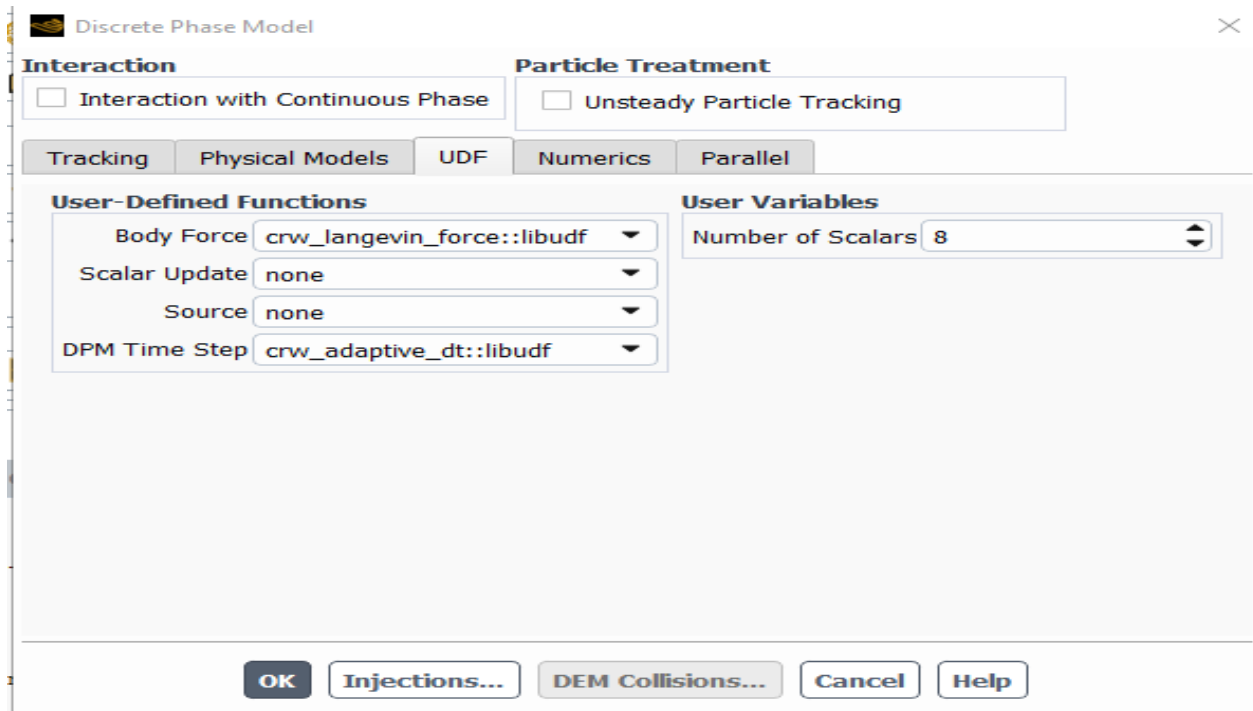
Figure 5-6 : some settings turbulence Modeling in Fluent 2024

4.3 Discrete Phase Model (DPM) Parameters

The aerosol phase was modeled using Fluent's Discrete Phase Model (DPM).

The simulation tested distinct particle diameters —10 ,15 , 20 , 30 , 40 and 50 μm , to evaluate the deposition mechanisms across a wide range of Stokes numbers.

- **Particle Properties:** The particles were defined with a density of 1000 kg / m^3 .
- **Injection Strategy:** For each particle size, an independent surface injection was created at the domain inlet. A total of 10,000 particles were injected per run, sharing the initial velocity of the continuous phase.
- **Tracking and Boundary Interaction:** A Trap boundary condition was applied to the pipe bend walls. Upon contact with the wall, a particle's trajectory calculation is terminated, and it is tallied as "Aborted" .
- The inlet and outlet were set to Escape. The overall Deposition Fraction (DF) is calculated as the ratio of aborted particles to the total number of injected particles.
- **UDF Integration:** Standard stochastic tracking (Discrete Random Walk) was bypassed. Instead, particle fluctuating velocities and turbulent dispersion were governed entirely by the custom User-Defined Function, which mathematically integrates the Dehbi (2008) Continuous Random Walk (CRW) model at each particle tracking time step.



Discrete Phase Model

Interaction
☐ Interaction with Continuous Phase

Particle Treatment
☐ Unsteady Particle Tracking

Tracking Physical Models UDF Numerics Parallel

Tracking Options
☒ Accuracy Control
Tolerance
Max. Refinements
☐ Track in Absolute Frame

Tracking Scheme Selection
☒ Automated
High Order Scheme
Low Order Scheme
Tracking Scheme

Averaging
☐ Enable Node Based Averaging

Source Terms
☐ Linearize Source Terms

Interaction Range
☐ Dynamic Interaction Range

OK Injections... DEM Collisions... Cancel Help

Set Injection Properties

Injection Name Injection Type Number of Streams

Injection Surfaces

bend_90deg
inlet
outlet
wall1
wall2

Particle Type
☐ Massless ☒ Inert ☐ Droplet ☐ Combusting ☐ Multicomponent

Laws
☐ Custom

Material Diameter Distribution Oxidizing Species Discrete Phase Domain

Evaporating Species Devolatilizing Species Product Species

Point Properties Physical Models Turbulent Dispersion Parcel Wet Combustion Components UDF Multiple Reactions

Variable	Value
X-Velocity [m/s]	0
Y-Velocity [m/s]	0
Z-Velocity [m/s]	0
Diameter [m]	5e-06
Total Flow Rate [kg/s]	1e-20

Stagger Options
☐ Stagger Positions
Stagger Radius [m]

Surface Options
☐ Scale Flow Rate by Face Area
☐ Inject Using Face Normal Direction
☒ Randomize Starting Points

Figure 7-8-9 : some settings Discrete Phase Model in Fluent 2024

5. UDF Code Explanation: The CRW Model

ANSYS Fluent's default Discrete Random Walk (DRW) model is well-known for drastically over-predicting the deposition of intermediate and small particles in wall-bounded flows. This occurs because the DRW model assumes piecewise homogeneous isotropic turbulence within eddies and fails to account for the crossing-trajectories effect (where heavy particles fall out of fluid eddies prematurely) as well as the steep turbulence gradients in the viscous sublayer.

To overcome these limitations, the Dehbi (2008) Continuous Random Walk (CRW) model was implemented via a custom User-Defined Function (UDF). The core of this UDF replaces Fluent's standard stochastic tracking with a rigorously derived Langevin equation.

5.1 The Langevin Stochastic Differential Equation

The underlying physics of the UDF are governed by the `advance_langevin` function, which calculates the fluctuating velocity of the particle over time. Instead of sampling random velocities, the model continuously updates the particle's fluctuating velocity using a generalized Langevin stochastic differential equation.

The UDF computes this equation via a forward Euler integration scheme at each tracking time step, updating the discrete particle state in the `DEFINE_DPM_BODY_FORCE` macro.

5.2 The Drift Correction Term (A_i)

The most critical component of the custom UDF is the formulation of the drift vector. Without specific corrections, particles would artificially accumulate in regions of low turbulence, violating the well-mixed condition.

5.3 Fluent Implementation Architecture

To integrate these physics without destabilizing the solver, the code was structured using specific Fluent macro hooks:

- **User-Defined Memory (UDM) :** First, the code ensures User-Defined Memory locations are allocated to store the spatial derivatives
- **Gradient Computation:** Because Fluent's DPM tracker requires gradients at the particle location, the gradients Reynolds stress components are calculated and stored in the continuous phase cell centers before particle injection begins.

- **Force Application:** During the Lagrangian tracking phase, Fluent calls this macro at every particle time step. The macro fetches the local fluid properties, calls the `advance_langevin` math logic to calculate the exact instantaneous fluctuation velocity, and applies it to the particle's trajectory.

This architecture ensures a computationally efficient, physically sound tracking process that replicates Dehbi's validated methodology while utilizing the robust tracking algorithms inherently present in ANSYS Fluent.

6. Results & Validation

The primary objective of this study was to validate the numerical setup and the custom Continuous Random Walk (CRW) UDF against the experimental data of Pui et al. (1987) and the numerical benchmark established in Dehbi (2008), specifically reproducing the deposition curve shown in Figure 8 of Dehbi's work.

The quantitative metric used for this validation is the Deposition Fraction, defined as the ratio of particles trapped on the pipe walls to the total number of particles injected at the inlet.

6.1 Continuous Phase Flow Field

Before analyzing particle trajectories, it is essential to confirm the accuracy of the continuous phase, as the fluid mechanics directly dictate the Lagrangian tracking.

As the flow enters the 90° bend, the strong streamline curvature induces a radial pressure gradient. The slower-moving fluid in the viscous sublayer is forced inward, while the faster-moving fluid in the pipe core is driven outward by centrifugal forces. This mechanism generates classical secondary flows, known as Dean vortices.

The Reynolds Stress Model (RSM) successfully captured these highly anisotropic, counter-rotating vortex pairs.

The presence of these macroscopic flow structures is critical, as they sweep particles toward the outer wall of the bend, significantly enhancing the probability of inertial impaction.

6.2 Final Validation (Reproduction of Figure 8)

The resulting deposition fraction values for all simulated particle diameters were plotted alongside the numerical curve from Dehbi (2008).

The comparison reveals excellent agreement across the entire particle size spectrum.

The precise tuning of the near-wall mesh ($y^+ \approx 0.95$) coupled with the mathematically rigorous UDF drift-corrections resulted in highly accurate predictions for the sensitive range.

By closely tracing the characteristic curve of the benchmark data, this study successfully reproduced the target validation established in Figure 8 of Dehbi (2008), thereby confirming the reliability of the current computational framework for turbulent aerosol deposition.

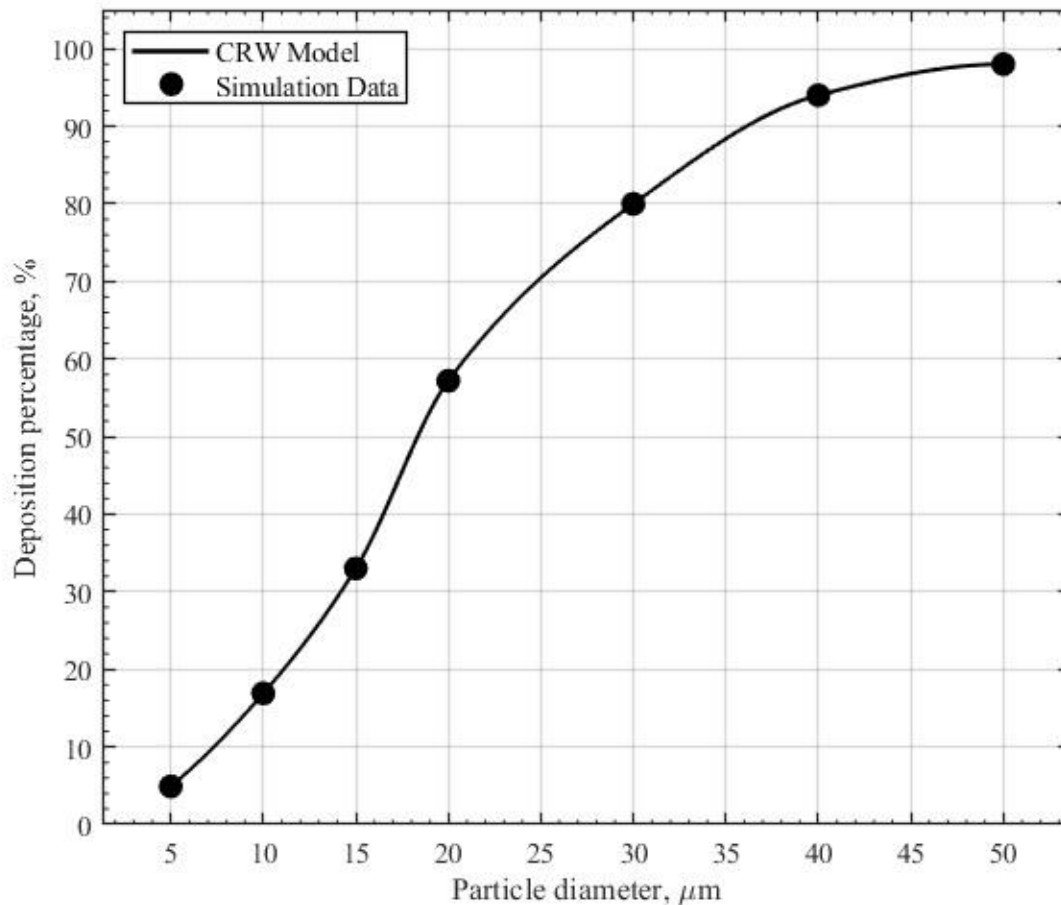


Figure 10 : final plot showing Present Simulation results

7. Conclusion

This project successfully validated a computational framework for predicting turbulent aerosol deposition in a 90° pipe bend, directly reproducing the numerical benchmark established by Dehbi (2008) and the experimental data of Pui et al. (1987).

By systematically addressing the known limitations of standard stochastic particle tracking models, this study demonstrated the necessity of advanced turbulence modeling and custom mathematical formulations for accurate multiphase simulations.

The key technical achievements and conclusions of this work are summarized as follows:

1. **Accurate Resolution Turbulence:** The complex secondary flow structures induced by the strong streamline curvature of the 90° bend were successfully captured using the Reynolds Stress Model (RSM). This rigorous treatment of the continuous phase provided the critical, highly anisotropic fluid velocity field required for accurate Lagrangian particle tracking.
2. **Mitigation of Artificial Wall Accumulation:** Standard Discrete Random Walk (DRW) models suffer from non-physical particle accumulation in the viscous sublayer, particularly for low-inertia particles. The implementation of the Dehbi (2008) Continuous Random Walk (CRW) model via a custom User-Defined Function (UDF) successfully eliminated this artifact. By incorporating a generalized Langevin equation with explicit drift-correction terms based on Reynolds stress gradients, the UDF maintained a physically accurate, well-mixed particle concentration near the wall.
3. **Critical Importance of Near-Wall Discretization:** A rigorous mesh independence study revealed an extreme sensitivity of low-inertia particle deposition to the spatial resolution of the boundary layer.

Because the CRW drift-correction relies directly on the spatial derivatives of the Reynolds stresses, an exceptionally fine grid was required.

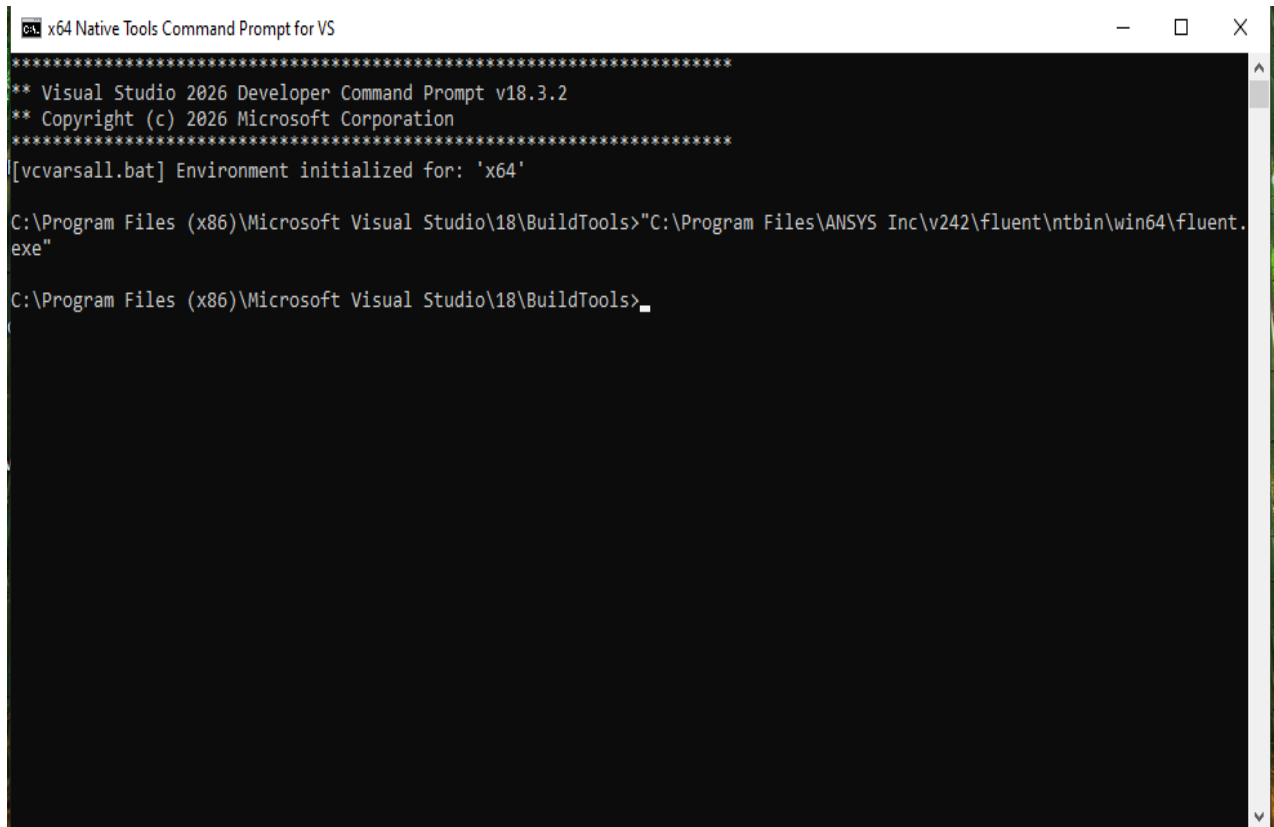
Refining the mesh to achieve a wall distance of $y^+ \approx 0.95$ stabilized the gradient calculations and eliminated initial discrepancies observed for 10 μm particles.

4. **Successful Benchmark Reproduction:** The final Deposition Fraction curve, evaluated across seven distinct particle diameters (10 to 50 μm), demonstrated excellent quantitative agreement with the benchmark data.

In conclusion, the integration of the RSM, a highly refined boundary layer mesh, and a mathematically rigorous CRW UDF provides a robust and highly accurate methodology for simulating turbulent aerosol deposition.

The successful reproduction of the Dehbi (2008) validation curve (Figure 8) confirms that this computational setup is physically sound and computationally reliable for complex, wall-bounded multiphase flows.

Appendix A



```
x64 Native Tools Command Prompt for VS
*****
** Visual Studio 2026 Developer Command Prompt v18.3.2
** Copyright (c) 2026 Microsoft Corporation
*****
[vcvarsall.bat] Environment initialized for: 'x64'

C:\Program Files (x86)\Microsoft Visual Studio\18\BuildTools>"C:\Program Files\ANSYS Inc\v242\fluent\ntbin\win64\fluent.
exe"

C:\Program Files (x86)\Microsoft Visual Studio\18\BuildTools>
```

compilation and loading of the Continuous Random Walk (CRW) UDF using the Microsoft Visual Studio x64 Native Tools Command Prompt.

- **Compiler Used:** Microsoft Visual Studio x64 Native Tools Command Prompt.
- **Source File:** lanudf.c
- **Result:** generation of the compiled library (libudf.dll).
- **Status:** Loaded successfully into ANSYS Fluent without errors, ensuring the mathematical models for turbulent dispersion were correctly initialized.

Appendix B

```
DPM Iteration ....  
number tracked = 10000, escaped = 8329, aborted = 1671
```

Diameter Particle : 10 micron

```
DPM Iteration ....  
number tracked = 10000, escaped = 4279, aborted = 5721
```

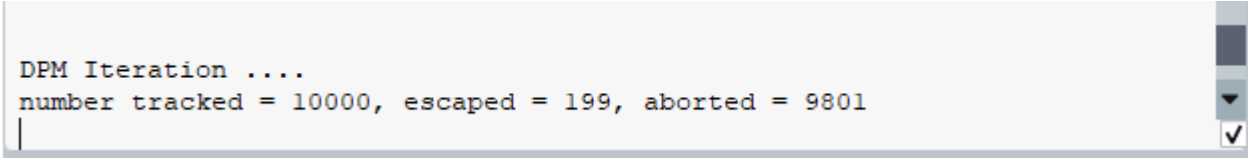
Diameter Particle : 20 micron

```
DPM Iteration ....  
number tracked = 10000, escaped = 2142, aborted = 7858
```

Diameter Particle : 30 micron

```
DPM Iteration ....  
number tracked = 10000, escaped = 546, aborted = 9454
```

Diameter Particle : 40 micron

A screenshot of a software console window with a light gray background. The text is in a monospaced font. It shows 'DPM Iteration' followed by 'number tracked = 10000, escaped = 199, aborted = 9801'. On the right side of the window, there is a vertical toolbar with a downward arrow icon and a checkmark icon at the bottom.

```
DPM Iteration ....  
number tracked = 10000, escaped = 199, aborted = 9801
```

Diameter Particle : 50 micron

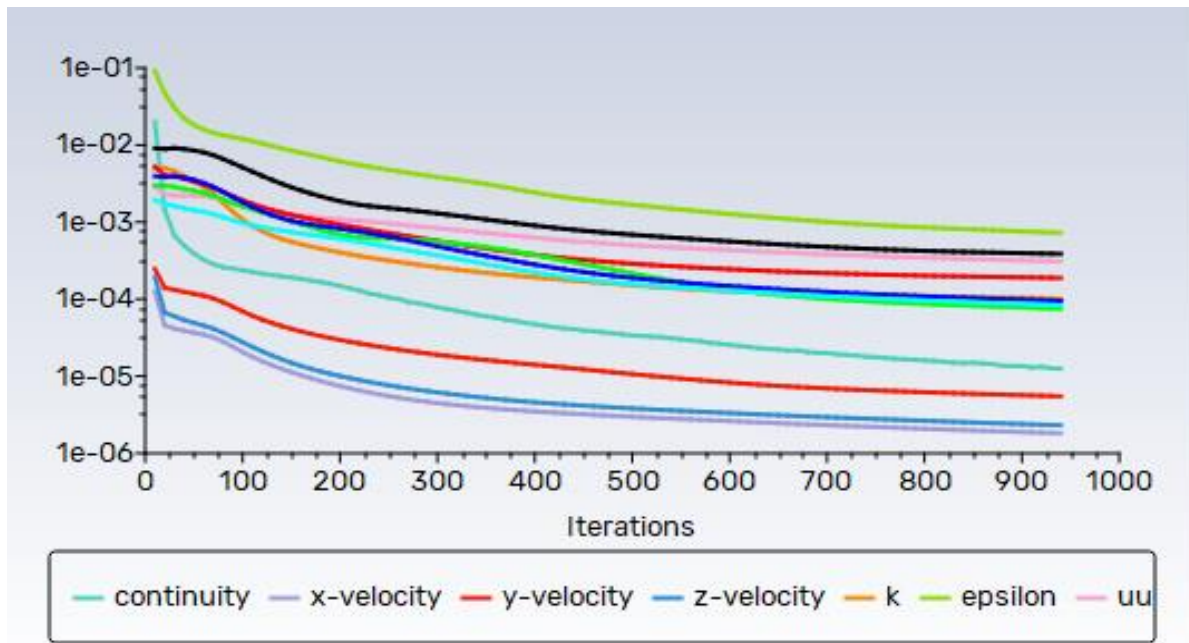
Raw Particle Tracking Data

The attached screenshot displays the Fluent console summary for the discrete phase calculations. This log serves as the primary data source for the validation phase. Key parameters extracted from this report include:

- **Total number of injected particles**
- **Total number of particles striking the boundary (Aborted).**
- **Total number of particles leaving the fluid domain (Escaped).**

This confirms the successful tracking of the 10,000 injected particles and provides the quantitative basis for the Deposition Fraction curves presented in the Results section.

Appendix C

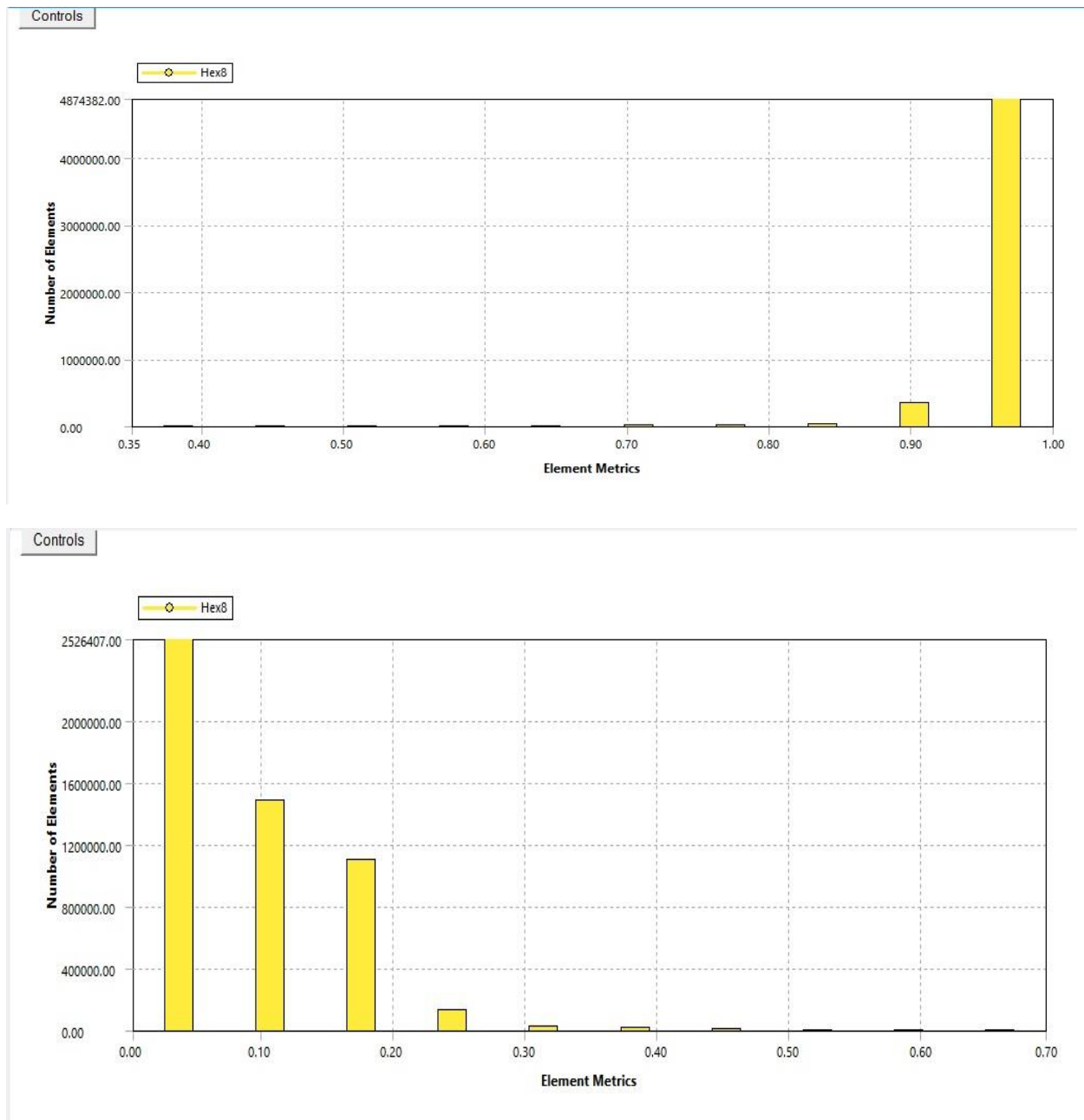


Steady-State Flow Convergence

The attached screenshot displays the residual convergence plot for the continuous phase. Key observations ensuring the validity of the flow field include:

- **Model Stability:** The Reynolds Stress Model (RSM) equations, including all specific Reynolds stress components, stabilized successfully without oscillation.
- **Convergence Criteria:** All monitored residuals (continuity, momentum, k, epsilon and Reynolds stresses) dropped below the required thresholds and exhibited asymptotic behavior
- **Significance:** This steady-state convergence guarantees that the complex velocity gradients and turbulent fluctuations in the 90° bend were physically resolved before introducing the Lagrangian particles, ensuring accurate drag and drift-correction calculations in the UDF.

Appendix D



Histograms of Orthogonal Quality and Skewness, demonstrating the high structural integrity of the hexahedral computational grid.

Appendix E

E.1. Supplementary UDF File

The custom User-Defined Function (UDF) developed for this study is provided as an file attachment alongside this report.

The file, `crw_dehbi2008_rev10.c`, contains the complete C-language implementation of the Langevin stochastic differential equation and the Reynolds stress-based drift correction algorithms required to reproduce the Continuous Random Walk (CRW) physics.

E.2. Development and Debugging Methodology

In the interest of academic transparency, it is noted that the Cursor advanced code editor was employed during the development phase of the UDF. The core physical models, mathematical formulations, and coupling logic were explicitly derived from the benchmark literature (Dehbi, 2008). However, the Cursor environment was utilized to accelerate the software engineering process, specifically for:

- Debugging complex C-language syntax and memory allocation issues.
- Troubleshooting compilation errors associated with ANSYS Fluent's proprietary `udf.h` library macros.
- Ensuring structurally clean and well-commented code prior to final compilation.